So basically we have covered statistics population Like population and sample population

Population: It is the entire part (1M)

Sample: It is the part of the whole(10K)

Q1. Company 1 M customers

What is population? 1M

Q2. The company selects 20k for survey?

What is sample for this population?

20k

Q3. Why don't we always take entire data/customers/population 1 M?

- To reduce cost
- To save time
- Resources
- Practical difficulty
- Accessibility

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Parameter

It is a numerical value that describes an entire population.
Population→ parameter
Example: A company has 10k employees.
AVG salary= $50,000
We have calculated the value using the entire population→ $50,000 is **Parameter**

Statistic

A statistic is a numerical value calculated from a sample

Sample→ Statistic

Example: Suppose we selected 500 random employees (sample)

Their avg salary is $51,000

$51,000 is a **statistic** because it was calculated from a sample.

Relationship

POPULATION→ PARAMETER

SAMPLE→ STATISTIC

Why do we need a statistic?

A company→5 M customers

AVG monthly spending

We don't know the actual population average

10k cx → calculated the average → 4,000→ Sample statistic

I have done 3 sampling→ The Statistic will not be same for different samples

We can use this data to learn about larger population.

This is foundation for **inferential statistics.**

Parameter vs Statistic

Parameter	Statistic
Describes population	Describes sample
Uses entire population	Uses sample
Usually fixed for a given population	Can vary between samples
Often unknown	Usually calculated from available sample data
Population Mean = μ	Sample Mean= $\bar{x}$
Population SD= σ	Sample SD= s

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Q1. What is the average salary of employees? 10K

Situation A: Census

All 10K employees are analyzed,

Average = 58,000 = parameter

Situation B: Sample

500 employees are selected

Average = 56,800 = Statistic

Census vs Sampling

Census:

A census collects data from every member of the population.

Sampling:

Sampling means collecting data from a selected subset of the population.

3000 std→ 300std

Census	Sampling
Everyone is studied	Only a subset is studied
More expensive	Less expensive
More time-consuming	Faster
Can require more resources	Requires fewer resources
Suitable for small populations	Often useful for large populations

Census over 1 M customers

Problems:

- Expensive
- Time Consuming
- Difficulty in management
- Accessibility

Sampling→ 10k customers→ Practical

When to use census?

- Population is small
- Data is easily available
- Complete information is required.
- Cost and time are manageable

When to use Sampling?

- Population is large
- Data collection is expensive
- Time is limited
- Testing every unit is impractical

➢ Sampling is better choice if population is large

➔ Sampling is more practical, but its quality depends on how the sample is collected(Quantiy wise)
S1 Average → 50000, S2 → 51000, S3→ 50500, S1/S2/S3 Avg→ 50500

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Variables & Observations:

What are we measuring about them

Variable:

A variable is characteristic or property that can take different values.

Age,Salary, Sales, Quantity = 5, Region,

Customer-id	Age	Sales
C101	22	₹5,000
C102	31	₹8,500
C103	27	₹6,200

Observation:

An observation is one individual record or a measurement in a dataset

Customer-id	Age	Sales
C101	22	₹5,000

The entire record is an observation.

Variable (column)	Observation(row/record)
Characteristic being measured	Individual recorded instance
Usually represented by a column	Usually represented by a row
Age, Salary, Sales	One customer's recorded values

Types of Data

---

1. Qualitative/Categorical Data
2. Quantitative/ Numerical Data

Qualitative Data

Qualitative data describes categories, qualities, or characteristics rather than numerical measurements.

Example:

- Age Group
- City
- Product Category

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- Gender
- Department
- Payment method
- Customer Feedback

Quantitative Data

Quantitative data represents numerical values that can be measured or counted

Example:

- Age
- Salary
- Sales
- Profit
- Quantity
- Height
- Weight

Qualitative	Quantitative
Categories / qualities	Numerical measurements
Usually describes "what type?"	Often answers "how much/how many?"
Department	Salary
Product Category	Sales
Payment Method	Quantity
Customer Feedback	Age

Ecommerce data

Customer-id	Region	Product	Quantity	Sales

Qualitative: region, Product
Quantitative: Quantity, Sale

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Discrete vs Continuous

This is subset of Quantitative data

Quantity→ discrete (1,10,15,0,5,6)

Roll_ Num→ continuous (001-010)

Age Groups→ 20-35,35-50,50-65

Age→ discrete

Discrete Data

Discrete data consists of countable values, usually whole numbers.

Example:

- No. of orders
- No. of Emp
- No. of Complatints
- No of defective products
- No. of children

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Continuous data

Continuous data can take any value within the range and us usually obtained during measurement.

Examples:

- Height
- Weight
- Temp
- Time
- Distance
- Sales amount

Time→ 32.5,32.57,32.573,32.5732

In time the precision is increasing to the right

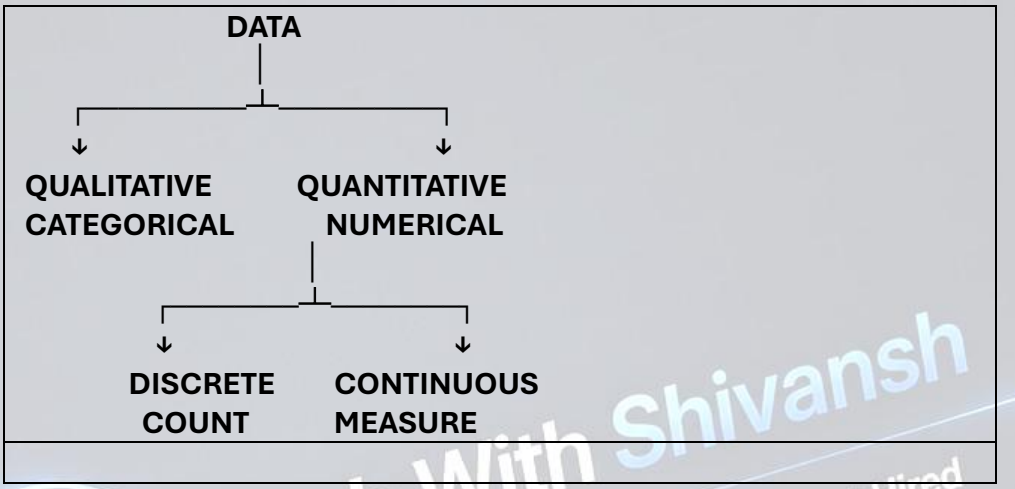
Discrete	Continuous
Counted	Measured
Usually whole numbers	Can include decimals
Number of orders	Delivery time
Number of employees	Employee height
Number of defects	Product weight

Interview trap:

Is salary discrete or continuous?

➔ Salary can be recorder at different levels of precision. In many practical datasets, salary is treated as **quantitative continuous variable** because monetary values can take many possible values within a range (decimals/whole number)

56540→26 days of work 25→ 54365.38



POPULATION (CENSUS) → PARAMETER

SAMPLE→ STATISTIC

1. What is a parameter?
2. What is statistic?
3. What is census?
4. Sampling?
5. Variable?
6. Observation?
7. Qualitative vs quantitative with an example→ product?
8. Employee-id quantitative?

➔ It is stored as number, but conceptually it is an identifier/categorical variable,→ not a meaningful quantitative measurement→ unique employees , unique cutomers

Variable	Classification
Customer Name	?
Age	?
City	?
Number of Orders	?
Monthly Income	?
Customer Rating	?
Product Category	?

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Variable	Classification
Delivery Time	?
Number of Complaints	?
Sales Amount	?

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